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Efficacy and metabolic influence of paliperidone ER, aripiprazole and ziprasidone to patients with first-episode schizophrenia through 52 weeks follow-up in China

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Background There are no direct comparisons of paliperidone extended-release (ER), aripiprazole and ziprasidone in efficacy and metabolic influence in patients with first-episode schizophrenia.
Objective The present study examined the efficacy and metabolic influence of paliperidone ER, aripiprazole and ziprasidone in patients with first-episode schizophrenia in China.
Methods Subjects were recruited from outpatient and 254 patients entered the trial. These patients received treatment randomly with paliperidone ER, aripiprazole and ziprasidone and were assessed at baseline, 13, 26 and 52 weeks, respectively with Positive and Negative Syndrome Scale (PANSS), 7-item Clinical Global Impressions-Severity (CGI-S), anthropometric (weight, body mass index and waist circumference) and metabolic (fasting blood glucose, HbA1c, cholesterol, high density lipoproteins (HDL), low density lipoproteins and triglycerides) measures.
Results A total of 203 patients completed the trial. Paliperidone group had significant greater reduction in PANSS than aripiprazole group and ziprasidone group from 13 weeks, although the a reduction in PANSS of each group was more than 20%. There was no difference in CGI-S among the three groups, and all three groups had a significant reduction from baseline in CGI-S. Aripiprazole group increased in weight and body mass index despite no statistical change in waist circumference. Other two groups showed no changes in anthropometric measure. At the end of the study, two glucose metabolic indices (fasting blood glucose and HbA1c) of aripiprazole group were significantly higher than that of baseline. In lipid metabolism, aripiprazole group reduced triglycerides significantly and had no changes in other indices. Paliperidone group reduced HDL and increased triglycerides despite no changes in glucose metabolism. Ziprasidone group also had no significant changes in glucose metabolism, but reduced cholesterol, low density lipoproteins and increased HDL. Furthermore, 22 subjects in three groups reached the diagnostic criteria of metabolic syndrome.
Conclusions Paliperidone ER, aripiprazole and ziprasidone are effective in treating first-episode schizophrenia, and the ranking of efficacy from high to low is paliperidone ER > aripiprazole > ziprasidone. Paliperidone ER can impair lipid metabolism potentially but had no influence on glucose metabolism. Aripiprazole can damage glucose metabolism and has little influence on lipid metabolism. Ziprasidone is considered an atypical antipsychotic with no evidence of harm to glucose and lipid metabolism. Copyright © 2012 John Wiley & Sons, Ltd.

KEY WORDS—schizophrenia; metabolism; paliperidone ER; aripiprazole; ziprasidone

INTRODUCTION

Schizophrenia, one of the most mysterious and costliest mental disorders, has caused heavy human suffering and societal expenditure (Mueser and McGurk, 2004). Patients with schizophrenia are likely to live about 12–15 fewer years than those without schizophrenia, with this mortality difference increasing in recent decades (Saha *et al.*, 2007). Suicide could explain part of this difference, the rest being due to cardiovascular disorder, smoking, obesity and little exercise (Van Os and Kapur, 2009). With the development of pharmaceuticals, atypical antipsychotics have gradually replaced first-generation antipsychotics. Atypical antipsychotics, now generally considered the standard of care for treatment of patients with schizophrenia (Mossman and Lehrer, 2000), appear to be effective against a wider range of symptoms and have a broader therapeutic window compared with conventional neuroleptics, although not in all contexts (Csernansky and Schuchart, 2002; Lieberman *et al.*, 2005). Although these medications are generally associated with a lower risk for extrapyramidal symptoms (EPS), the Clinical Antipsychotic Trials of Intervention Effectiveness study found that there is still 4–8% rate of EPS associated with their use (Lieberman *et al.*, 2005).

In addition, atypical antipsychotics may result in adverse metabolic effects, potentially contributing to an increased risk for diabetes mellitus and cardiovascular

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disease (Shirzadi and Ghaemi, 2006; Newcomer, 2007). Research evidence shows that atypical antipsychotics can impact metabolism of patients including abdominal obesity, dyslipidaemia, hyperglycemia, and fasting plasma glucose.

Among these atypical antipsychotics, three medications have unique features: aripiprazole, ziprasidone and paliperidone. The antipsychotic effect of ziprasidone is proposed to be primarily via D_2 and serotonin (5-HT_2) eceptor antagonism (Miyamoto *et al*., 2005). Aripiprazole is a partial agonist at D_2 receptors; thus, it is an antagonist in the presence of high levels of endogenous dopamine and, conversely, acts as an agonist when minimal dopamine is present. Aripiprazole is also a partial agonist at 5-HT_{1A} receptors that may contribute to improvements in anxiety, depression, negative symptoms and lower incidence of extrapyramidal side effects (Wood and Reavill, 2007). Paliperidone extended-release (ER), the most advanced atypical antipsychotic, was formulated to combine the active metabolite of risperidone (9-OH risperidone) with an osmotic controlled-release oral delivery system that minimizes peak to trough fluctuations (Conley *et al*., 2006). Paliperidone's mechanism of action is believed to result from its high antagonist activity at the D_2 and 5-HT_{2A} receptors. It also has additional antagonistic affinity for the histaminic H1 and adrenergic α-1 and α-2 receptors. Paliperidone ER is the first and only antipsychotic that is not metabolized by the P-450 isoenzyme system and is only minimally metabolized by other pathways in the liver. Thus, paliperidone would not be expected to be subject to drug–drug interactions mediated through the P-450 system, and its dosing would not be expected to differ between slow and rapid metabolizers (Karlsson *et al*., 2006).

Current evidence suggests that aripiprazole and ziprasidone have most weak metabolic effect (De Hert *et al*., 2006; Haupt, 2006; Haddad and Sharma, 2007). However, few studies of long-term focused on the metabolic effect of paliperidone ER, whereas some conclusions presented that paliperidone ER had little influence on weight gain and more influence on prolactin (Emsley *et al*., 2008; Hough *et al*., 2009). Although past studies conclude that all three drugs had little impact on glucose and lipid metabolism, no research has compared the three medications directly. Hence, this prospective study expects to find detailed influences of paliperidone ER, aripiprazole and ziprasidone on metabolism of patients with schizophrenia as well as efficacy through a year follow-up in a Chinese a setting.

METHOD

Subjects

From March 2010, participants were recruited from the outpatient clinic of the Fourth People's Hospital of Chengdu. Subjects between the ages of 18 and 65 years with first-episode of schizophrenia were suitable for the study. Patients were excluded on the basis of current substance abuse; diabetes mellitus; thyroid disease; pregnancy; significant medical illness including severe cardiovascular, hepatic or renal disease; or unstable psychiatric illness. Eligibility was determined by interview and a medical record review for history and recent laboratory values. Patients treated with medications known to affect metabolism were also excluded. A urine pregnancy test was performed prior to the study for female subjects of childbearing potential. All patients were diagnosed with schizophrenia according to *Diagnostic and Statistical Manual of Mental Disorders*, *Fourth Edition* criteria. All patients gave written informed consent to the procedures of the study, which was approved by the local research ethics committee.

Procedure

A total 254 subjects received treatment with paliperidone (Invega® ER Tablet, 6 mg), aripiprazole (Abilify® Tablet, 5 mg) and ziprasidone (Geodon® Capsule, 20 mg) randomly. Paliperidone group had 81 subjects, aripiprazole group had 90 subjects and ziprasidone group had 83 subjects. Subjects were instructed to take medication regularly as prescribed and did not eat anything at the testing day, in which blood samples were taken. This is an open-label study despite of randomization. Firstly, each medication was labeled with number: aripiprazole as 1; ziprasidone as 2 and paliperidone as 3. Each patient identifying with first-episode schizophrenia would get a card with number 1–3 randomly from a nurse working in outpatient. The psychiatrists in outpatient clinics distributed medications to patients as the number corresponding to the medication. Subjects received 52 weeks follow-up. At the end, 203 patients completed the trial (79.9%) (Figure 1). The results of anthropometric measure including weight, body mass index (BMI), waist circumference and metabolic measure including fasting blood glucose, hemoglobin A1c (HbA1c), cholesterol, high density lipoproteins (HDL), low density lipoproteins (LDL) and triglycerides were recorded at baseline, 13, 26 and 52 weeks, respectively. Clinical status was assessed with the tools of the Positive and Negative Symptom Scale (PANSS) and the 7-item Clinical Global Impressions-Severity (CGI-S). In addition,

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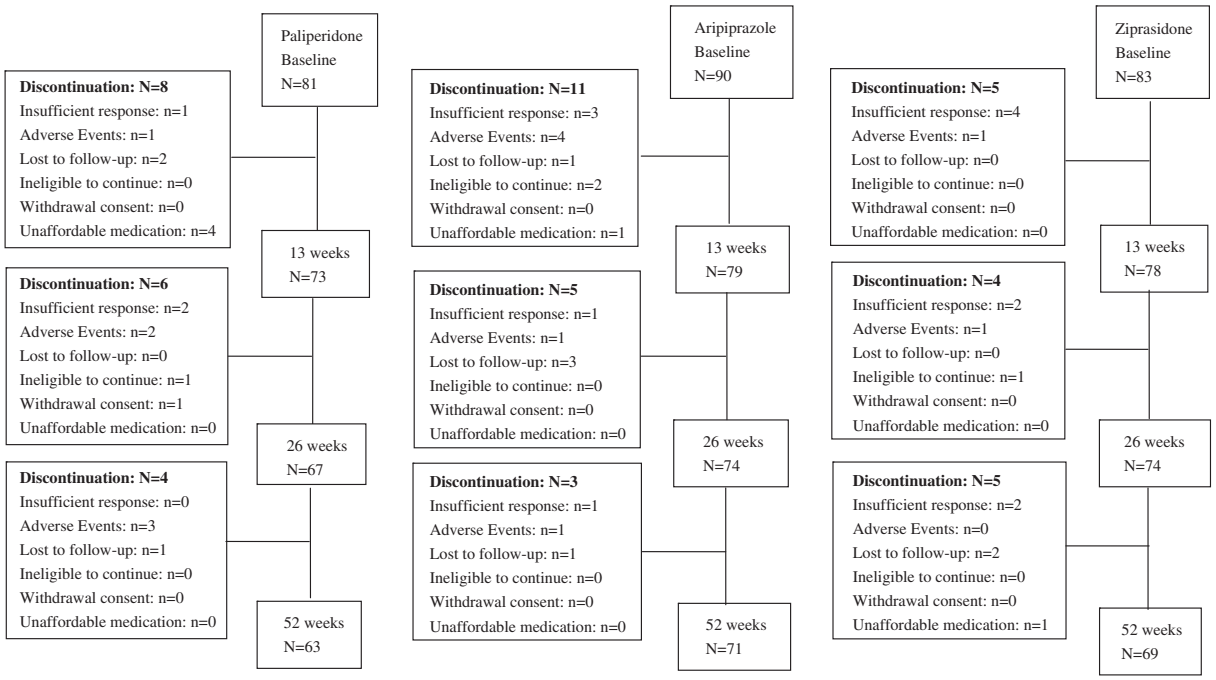

Figure 1. Patients disposition. A total of 254 subjects successfully entered the trial and 203 subjects completed the trial (79.9%). *In china, a pill of Paliperidone (6 mg) was about 7.5 dollars; a pill of Aripiprazole (5 mg) was about 2 dollars; and a pill of Ziprasidone (20 mg) was about 3 dollars

safety and tolerability assessments were carried out, which included adverse events, the EPS, vital signs, electrocardiograms and physical examinations. EPS was evaluated by the EPS Rating Scale.

Statistical analysis

Statistical analyses were conducted with SPSS 13.0 (SPSS Inc, Chicago, IL, USA). Descriptive statistics are represented as mean (Standard Deviation). Categorical demographic variables were compared between groups using the *Chi-square* test. Continuous demographic variables among three groups were compared using analysis of variance. *Post hoc* test was used if the overall group effect F test p value was less than 0.05. When comparing mean changes from baseline, t test was used. A p value less than 0.05 was used to test for statistical significance.

RESULTS

Demographic characteristics

A trial of 51 subjects discontinued for the following reasons: insufficient response (16), adverse events (14), lost to follow-up (10), ineligible to continue (4), withdrawal consent (1) and unaffordable medication (6). The rest of 203 subjects completed the study: paliperidone group (63), aripiprazole group (71) and ziprasidone group (69) (Figure 1). For the entire 203 subjects, the average age was 26.4 (7.5) and 124 (61%) were male. The three groups did not differ in age, gender, family history of metabolic disturbance and duration of illness (Table 1). Compared with discontinued proportion of each two groups, the results of odds ratio showed that paliperidone group (22.2%) versus aripiprazole group (21.1%) was 1.06 (95% CI, 0.51 to 2.21); paliperidone group (22.2%) versus ziprasidone group (16.8%) was 1.41 (95% CI, 0.65 to 3.06); aripiprazole group (21.1%) versus ziprasidone group (16.8%) was 1.32 (95% CI, 0.61 to 2.84). There was no significant difference between groups.

PANSS and CGI-S measures

There were no significant differences in PANSS and CGI-S in the three groups at baseline. From 13 to 52 weeks, there were differences among the three groups in PANSS. Post hoc tests showed that the paliperidone group had significantly lower scores than the aripiprazole group ($p = 0.011$, 95% CI, 0.91 to 2.98) and the ziprasidone group ($p = 0.024$, 95% CI, 0.55 to 7.05). Contrary to PANSS, there were no differences among groups in CGI-S throughout the trial (Table 2). However, each group had significant difference in changes of PANSS and CGI-S from baseline at all phases ($p < 0.05$). Further, there were significant differences among the three groups in mean

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Table 1. Demographics of participants receiving antipsychotics

Demographics*	Paliperidone ($n=63$)	Aripiprazole ($n=71$)	Ziprasidone ($n=69$)	F/χ^2	p-value
Gender, no. (%) of patients				1.18	0.555
Male	35(55.6)	45 (63.4)	44 (63.7)		
Female	28(44.4)	26 (36.5)	25 (36.3)		
Age, mean (SD)	27.1 (7.4)	25.7 (6.9)	26.3 (7.2)	1.97	0.158
Diagnose of hypertension, no. (%) of patients				0.32	0.852
Yes	8(12.7)	11 (15.5)	11 (15.9)		
No	55(87.3)	60 (84.5)	58 (84.1)		
Duration of illness, month and mean (SD)	2.5 (1.4)	2.1 (1.1)	2.3 (1.0)	1.85	0.176

*Age: in years.
Hypertension: flagged as present when the subject was taking antihypertensive medication or had measured blood pressure >140/90 mmHg.

Table 2. Analysis of PANSS and CGI-S in three groups at different phases

Parameter*	Paliperidone ($n=63$)	Aripiprazole ($n=71$)	Ziprasidone ($n=69$)	F	p-value
Baseline					
PANSS	87.1 (12.8)	89.8 (14.5)	88.0 (11.6)	1.03	0.369
CGI-S	4.6 (2.2)	4.5 (1.9)	4.7 (2.1)	0.97	0.405
13 weeks later					
PANSS	59.6 (14.1)[a]	74.4 (13.7)[b]	73.6 (9.5)[b]	5.97	0.004
CGI-S	3.5 (1.7)	3.6 (1.3)	3.5 (1.5)	0.22	0.802
26 weeks later					
PANSS	55.1 (10.4)[a]	71.4 (13.7)[b]	72.9 (8.5)[b]	6.54	0.002
CGI-S	3.3 (1.6)	3.5 (1.4)	3.3 (1.8)	2.14	0.137
52 weeks later					
PANSS	54.9 (10.6)[a]	68.6 (9.3)[b]	69.3 (9.7)[b]	4.92	0.012
CGI-S	3.1 (1.1)	3.3 (1.2)	2.3 (1.4)	3.09	0.062

PANSS, Positive and Negative Symptom Scale; CGI-S, 7-item Clinical GlobalImpressions-Severity.
*Values are expressed as mean (SD) unless specially expressed.
[a,b]Significant difference between groups at $p < 0.05$.

changes in PANSS from baseline at 13 weeks ($F = 4.91$, $p = 0.025$), 26 weeks ($F = 5.45$, $p = 0.012$) and 52 weeks ($F = 5.13$, $p = 0.018$). As for CGI-S, there were no differences between the three groups (Figure 2).

Anthropometric measures

There were no differences in anthropometric measures at baseline. After 13 weeks, weight of aripiprazole group was statistically higher than that of other two groups. At 26 and 52 weeks, weight and BMI of aripiprazole group were higher than that of other two groups (Table 3).

Metabolic measures

At the baseline, there were no differences in six indices. At the end of the study, two glucose metabolic indices (fasting blood glucose and HbA1c) of aripiprazole group were significantly higher than that of baseline. In lipid

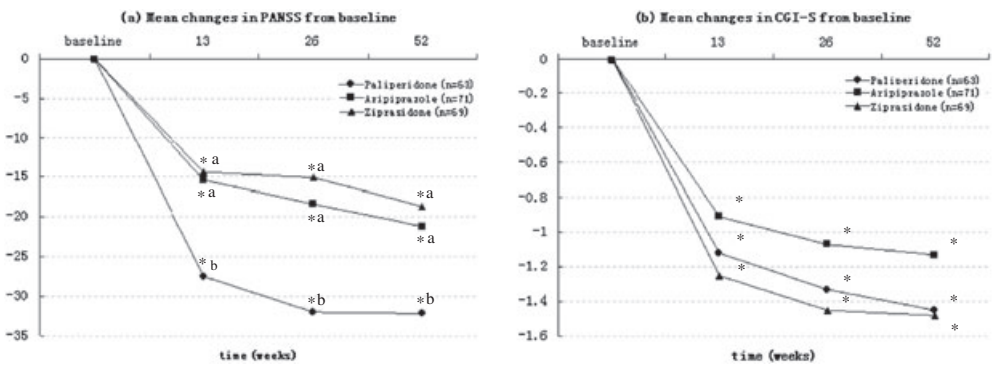

Figure 2. Mean changes in PANSS and CGI-S from baseline throughout the trial. (a) Each group had significant mean changes in PANSS from baseline. Paliperidone group had a significantly greater reduction than aripiprazole group and ziprasidone group at 13, 26 and 52 weeks, respectively ($p < 0.05$). (b) Each group had significant mean changes in CGI-S from baseline. There were no differences among the three groups from 13 to 52 weeks. a, b Significant difference between groups at $p < 0.05$. * Significant change from baseline at $p < 0.05$

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Table 3. Comparison of anthropometric results among groups at different phases

Parameter*	Paliperidone ($n = 63$)	Aripiprazole ($n = 71$)	Ziprasidone ($n = 69$)	F	p-value
Baseline	Mean dose, mg: 5.2 (2.1)	Mean dose, mg: 5.6 (4.3)	Mean dose, mg: 25.4 (12.4)		
Weight	53.4 (6.3)	54.3 (7.2)	54.7 (6.8)	0.38	0.645
BMI	21.5 (3.2)	22.4 (3.7)	21.9 (3.4)	0.97	0.322
WC	69.5 (12.4)	67.3 (14.6)	71.5 (13.8)	2.11	0.248
13 weeks later	Mean dose, mg: 6.4 (5.3)	Mean dose, mg: 12.4 (6.5)	Mean dose, mg: 36.4 (15.6)		
Weight	**52.3 (6.4)**[b]	**57.6 (7.7)**[a]	**52.7 (7.1)**[b]	**3.02**	**0.039**
BMI	21.2 (3.1)	23.7 (4.1)	20.2 (3.3)	1.94	0.115
WC	68.7 (15.3)	70.5 (16.2)	70.4 (15.8)	0.87	0.331
26 weeks later	Mean dose, mg: 6.4 (5.3)	Mean dose, mg: 16.8 (7.3)	Mean dose, mg: 68.2 (27.1)		
Weight	**53.1 (7.2)**[b]	**58.8 (8.5)**[a]	**51.9 (7.2)**[b]	**3.33**	**0.034**
BMI	**21.9 (5.2)**[b]	**24.3 (5.8)**[a]	**20.6 (5.1)**[b]	**4.12**	**0.027**
WC	68.4 (15.5)	71.3 (16.6)	69.7 (16.3)	2.26	0.178
52 weeks later	Mean dose, mg: 6.4 (5.3)	Mean dose, mg: 14.5 (8.2)	Mean dose, mg: 65.3 (28.4)		
Weight	**53.6 (7.4)**[b]	**57.4 (8.2)**[a]	**50.5 (6.9)**[b]	**3.27**	**0.037**
BMI	**21.5 (5.4)**[b]	**24.5 (5.9)**[a]	**20.3 (5.2)**[b]	**4.62**	**0.015**
WC	68.5 (15.6)	71.6 (17.6)	70.3 (16.7)	2.54	0.126

*Values are expressed as mean (SD) unless specially expressed.
WC, waist circumference; BMI, body mass index.
Weight: with light clothing and without shoes, in kilogram.
Body mass index: obtained from weight in kilogram divided by height in meters squared.
Waist circumference: measured in centimeters at a point midway between the iliac crest and the lower edge of the ribcage with the subject standing, in centimeter.
[a,b]Significant difference between groups at $p < 0.05$.

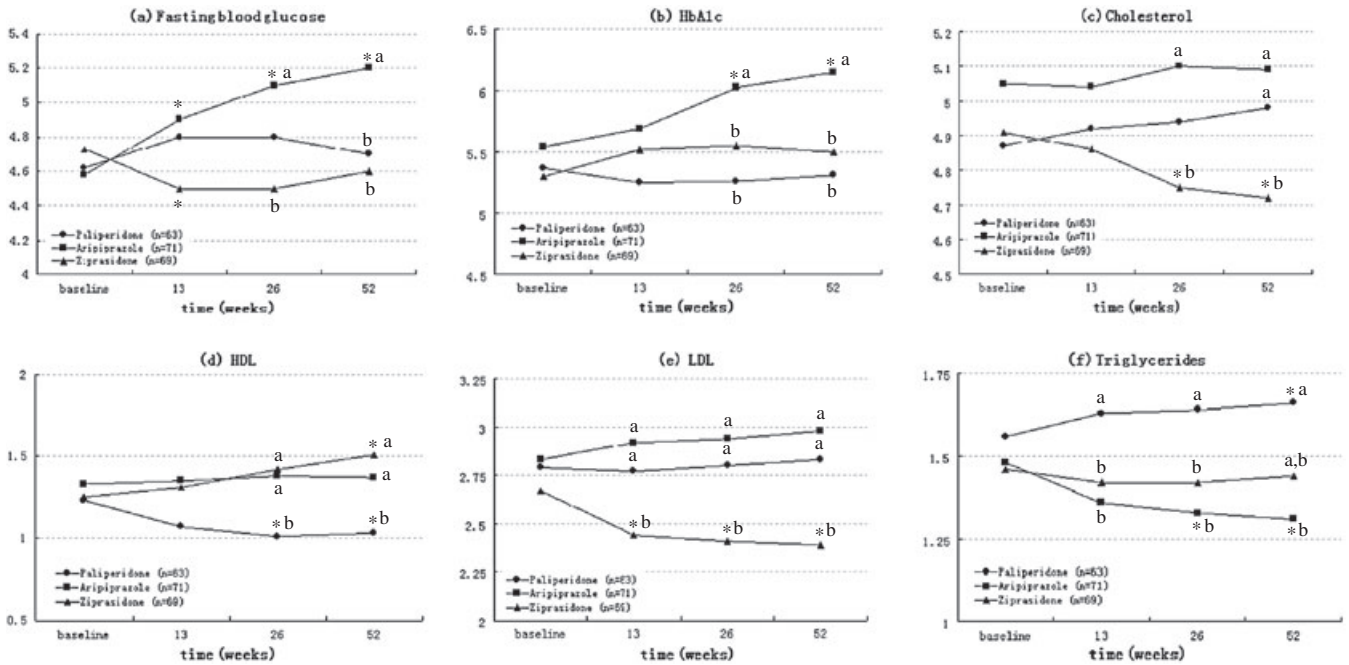

Figure 3. Analysis of metabolic results in groups at different phases. At the baseline, six indices of metabolic measures showed no differences in three groups. (a) At 13 weeks, the aripiprazole group and ziprasidone group showed differences from baseline. After 26 and 52 weeks, only the aripiprazole group showed difference from baseline. Moreover, the aripiprazole group showed difference with the paliperidone group and ziprasidone group at 52 weeks. (b) At 26 and 52 weeks, only the aripiprazole group showed difference from baseline and showed differences with the paliperidone group and ziprasidone group. (c) At 26 and 52 weeks, only the ziprasidone group showed difference from baseline. At 26 weeks, the ziprasidone group showed difference with the aripiprazole group. At 52 weeks, the ziprasidone group showed difference with the aripiprazole group and paliperidone group. (d) At 26 weeks, the paliperidone group showed difference from baseline and showed differences with the aripiprazole group and ziprasidone group. At 52 weeks, the paliperidone group and ziprasidone group showed differences from baseline and the paliperidone group showed differences with the aripiprazole group and ziprasidone group. (e) At 13, 26 and 52 weeks, the ziprasidone group showed difference from baseline and showed differences with the paliperidone group and aripiprazole group. (f) At 26 weeks, only the ziprasidone group showed difference from baseline; and at 52 weeks, the ziprasidone group and paliperidone group showed differences from baseline. At 13 and 26 weeks, the paliperidone group showed differences with the aripiprazole group and ziprasidone group. At 52 weeks, both groups showed difference. * Significant change from baseline at $p < 0.05$. a, b Significant difference between groups at $p < 0.05$

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metabolism, aripiprazole group reduced triglycerides significantly and had no changes in other indices. Paliperidone group reduced HDL and increased triglycerides despite no changes in glucose metabolism. Ziprasidone group also had no significant changes in glucose metabolism, but reduced cholesterol, LDL and increased HDL (Figure 3).

Summary of abnormal laboratory results at the end of the trial showed that there were 19 subjects in fasting blood glucose, 15 subjects in HbA1c, eight subjects in cholesterol, 27 subjects in HDL, 34 subjects in LDL (12 in abnormal high and 15 in abnormal low) and 14 subjects in triglycerides (Table 4). Furthermore, 22 subjects reached the diagnostic criteria of metabolic syndrome (Table 5) (International Diabetes Federation, 2005).

Table 5. Diagnose of metabolic syndrome* at the end of the trial

Paliperidone ($n=63$)	Aripiprazole ($n=71$)	Ziprasidone ($n=69$)	Total ($n=203$)
5 (8%)	11 (15.5%)	6 (9%)	22 (11%)

*New International Diabetes Federation criteria for metabolic syndrome (2006).
Central obesity: defined as waist circumference ≥90 cm for men and ≥80 cm for women, plus any two of the following four factors:

1. raised serum triglyceride level: ≥150 mg dL^{-1} (1.7 mmol L^{-1}) or specific treatment for this lipid abnormality;
2. reduced serum HDL-cholesterol level: <40 mg dL^{-1} (1.03 mmol L^{-1}) in male subjects and <50 mg dL^{-1} (1.29 mmol L^{-1}) in female subjects or specific treatment for this lipid abnormality;
3. raised blood pressure: systolic BP ≥130 mmHg or diastolic BP ≥85 mmHg or treatment of previously diagnosed hypertension; and
4. raised fasting plasma glucose: ≥100 mg dL^{-1} (5.6 mmol L^{-1}) or previously diagnosed type 2 diabetes.

DISCUSSION

"Atypical" antipsychotic agents are used to treat the symptoms of schizophrenia and bipolar disorder. Atypical antipsychotics may also treat negative symptoms and improve cognitive functioning. In general, atypical antipsychotics produce antipsychotic responses with fewer acute extrapyramidal side effects than "conventional" antipsychotic drugs (Mossman and Lehrer, 2000).

The comparisons of the discontinuation rates showed that the ziprasidone group had lower discontinuation rate than the paliperidone group and aripiprazole group. Among the reasons of discontinuation, lack of effect and adverse events were the major concerns (31.3%, 27.5%, respectively). However, a special reason for discontinuation was important in Chinese setting, which was unaffordable medication (11.8%). Despite partial reimbursement of medication expenditure from Human Resources and Social Security Bureau, the rest part of self-paying was also a burden for some families. In China, the reimbursement of medication expenditure in outpatient from Human Resources and Social Security Bureau was different according to registered permanent residence of patients. Patients of registered permanent residence in cities could get 90% and higher reimbursement of medication expenditure, whereas patients of registered permanent residence in countries could only get about 50% reimbursement of medication expenditure. But patients without paid medical insurance fee regularly (monthly) have to pay the medication without any reimbursement. Therefore, country patients and self-paying patients were the most likely to discontinue medication.

Table 4. Summary of abnormal laboratory results at the end of the trial

Parameter	Paliperidone ($n=63$)	Aripiprazole ($n=71$)	Ziprasidone ($n=69$)	Total ($n=203$)
Fasting blood glucose				
Abnormal high >6.1 mmol L^{-1}	3 (4.7%)	14 (19.7%)	2 (2.9%)	19 (9.4%)
HbA1c (%)				
Abnormal high >6.5	4 (6.3%)	9 (12.7%)	2 (2.9%)	15 (7.4%)
Cholesterol				
Abnormal high >6.47 mmol L^{-1}	2 (3.2%)	5 (7.0%)	1 (1.4%)	8 (3.9%)
HDL				
Abnormally low <0.90 mmol L^{-1}	8 (12.7%)	15 (21.1%)	4 (5.8%)	27 (13.3%)
LDL				
Abnormal high >3.1 mmol L^{-1}	4 (6.3%)	7 (9.9%)	1 (1.4%)	12 (5.9%)
Abnormal low <2.7 mmol L^{-1}	5 (7.9%)	3 (4.2%)	7 (10.1%)	15 (7.4%)
Triglycerides				
Abnormal high >4.5 mmol L^{-1}	8 (12.7%)	4 (5.6%)	2 (2.9%)	14 (6.9%)

HbA1c, hemoglobin A1c; HDL, high density lipoproteins; LDL, low density lipoproteins.

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Up to now, no research has compared the efficacy of paliperidone ER, aripiprazole and ziprasidone directly. Some studies concluded that only four second-generation antipsychotics (clozapine, olanzapine, amisulpride and risperidone) were superior to first-generation antipsychotics in overall efficacy, whereas others, such as aripiprazole, quetiapine, and ziprasidone, were not (Leucht *et al*., 2009). A meta-analysis held that aripiprazole and ziprasidone showed smaller CGI-C score improvements than haloperidol (Lepping *et al*., 2011). As for paliperidone, three multi-centre, double-blind and the placebo-controlled open-label 6-week studies supported its efficacy (Davidson *et al*., 2007; Kane *et al*., 2007; Marder *et al*., 2007). A 52-week study including 1083 patients concluded that paliperidone ER can maintain improvements in symptoms and functioning (Emsley *et al*., 2008). In the present study, reductions in PANSS of the three groups at the end of the trial were more than 20%. Furthermore, the result showed that mean changes in PANSS from baseline of paliperidone group were statistically significantly greater than aripiprazole group and ziprasidone group, which indicated that paliperidone ER had better efficacy than aripiprazole and ziprasidone. Another advantage of paliperidone was taking medication conveniently. A total of 89.3% patients of the paliperidone group in this study took only one tablet each day, which meant better compliance for patients despite a relatively higher price.

Antipsychotic-induced weight gain is a complex side-effect with multiple causal mechanisms such as energy expenditure and intake (Gothelf *et al*., 2002), receptors and neurotransmitters (Wirshing *et al*., 1999; Reynolds *et al*., 2006; Kim *et al*., 2007a), pharmacogenetics (Reynolds *et al*., 2002; Buckland *et al*., 2005; Muller and Kennedy, 2006; Mulder *et al*., 2007), leptin (Bromel *et al*., 1998; Kraus *et al*., 1999; Melkersson *et al*., 2000; Hagg *et al*., 2001), ghrelin (Popovic *et al*., 2007), orexins (Fadel *et al*., 2002) and metabolism of glucose and lipids (Casey *et al*., 2004; Henderson, 2005; Holt and Peveler, 2006; Newcomer and Haupt, 2006). It is clear that two receptors associated with weight gain are the Serotonin 2C (5-HT_{2C}) and histamine 1 (H_1) receptors. Ziprasidone is a potent 5-HT_{1a} receptor agonist and a potent 5-HT_{2C} antagonist. Its slight effect on weight may be explained by its synaptic reuptake inhibition of serotonin and norepinephrine (Virk *et al*., 2004). Aripiprazole, in contrast, has a unique mechanism of action with partial D_2 and 5-HT_{1a} agonism and 5-HT_{2a} antagonism. Its low weight gain potential is attributed to its moderate affinity to H_1 receptors and lack of 5-HT_{2C} antagonism (Potkin *et al*., 2003; McQuade *et al*., 2004). Paliperidone has high antagonist activity at 5-HT_{2a} and H_1 receptors (Karlsson *et al*., 2006). An influential meta-analysis compared nine second-generation antipsychotics by Leucht *et al*. showed that all the second-generation antipsychotics, with the exception of ziprasidone and aripiprazole, caused more weight gain than haloperidol, to varying degrees, but not more than is seen with the low-potency first-generation antipsychotics (Leucht *et al*., 2009). Two studies of paliperidone showed that the influence on weight was relative to dose (Emsley *et al*., 2008; Meltzer *et al*., 2008). In this study, paliperidone and ziprasidone had no effect on weight gain, but aripiprazole increased weight and BMI of the patients. The results of waist circumference supplied a cue to explain this expression. There was no difference among the three groups in waist circumference measure, which indicated that weight gain did not focus on viscera and distributed to other positions such as limbs, trunk or buttocks. This could be caused by the relatively obvious sedation effect of aripiprazole, which resulted in hypoactivity.

Metabolic disturbance is an important side effect generated by atypical antipsychotics, which, in turn, could impair the compliance of patients. Many patients discontinued treatment because of metabolic disturbance, which could result in other diseases such as cardiovascular condition and diabetes mellitus. Atypical antipsychotics are more likely to produce this effect than the first-generation agents, but both contribute to impaired blood glucose mainly through insulin resistance (Lambert and Chapman, 2004). Dyslipidemia, on the other hand, also causes insulin resistance, which increases adipose tissue deposition (Wang *et al*., 2006). Thus, there exists a complex relationship between adiposity, blood glucose and serum lipids in relation to anti-psychotics. For atypical antipsychotics, effects on metabolism are different. Three separate open-label 6-week multi-centric studies showed that switching to ziprasidone was an effective strategy in individuals on olanzapine and produced reductions in lipids and weight (Weiden *et al*., 2003). Aripiprazole had been evaluated in randomized controlled trials with positive effects on metabolism of patients (Casey *et al*., 2003; Koponen and Larmo, 2003; Gupta *et al*., 2004; Lin and Chen, 2006; Kim *et al*., 2007b). For paliperidone, most authors of studies had affiliations with Janssen-Cilag or Johnson & Johnson or were supported by the companies (Citrome, 2007; Dlugosz and Nasrallah, 2007; Janicak *et al*., 2007; Spina and Cavallaro, 2007; Yang and Plosker, 2007; Birnbaum and Sharif, 2008; Nussbaum and Stroup, 2008; Jones *et al*., 2010). There are no studies directly comparing paliperidone, aripiprazole and ziprasidone in metabolic effect. The results of this

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study showed different results from previous studies. In this study, aripiprazole demonstrated that it could impair glucose metabolism, which was contrary to previous studies (Lambert *et al*., 2006; Hasnain *et al*., 2009; Patel *et al*., 2009). But in lipids metabolism, aripiprazole could reduce triglycerides and had no influences on other lipid metabolic indices. So, the conclusion of aripiprazole was paradoxical for it had little influence on lipid metabolism, but could damage glucose metabolism. Moreover, at the end of the trial, the subjects of aripiprazole had the most high proportion (15.5%) reaching the diagnostic criteria of metabolic syndrome, which increased controversy. Ziprasidone reduced cholesterol, LDL and enhanced HDL at the end of trial and had no impact on glucose metabolism, which indicated that ziprasidone could be a good choice for patients at risk of metabolic disturbance or be an alteration for patients with impaired metabolism produced by other antipsychotics. Paliperidone ER, the new atypical antipsychotic, had no influence on glucose metabolism, but reduced HDL and increased triglycerides when the study finished. Therefore, it should be conservative to consider that paliperidone ER could be suitable for patients with metabolic disturbance, which was not consistent with the previous studies (Davidson *et al*., 2007; Kane *et al*., 2007; Marder *et al*., 2007). In this study, only ziprasidone could be deemed an atypical antipsychotic with no harm to metabolism. Aripiprazole and paliperidone should be researched further.

Despite of different conclusions in this study, a different method compared with former studies should be noticed. In this study, the subjects were patients with first-episode schizophrenia and many previous studies focused on switching to other antipsychotics. In other words, the subjects in those studies were not in first-episode schizophrenia. Some studies reported that drug-naive schizophrenia patients had higher baseline values of weight and visceral fat distribution, as well as laboratory evidence of impaired glucose and lipid metabolism (Ryan *et al*., 2004; Spelman *et al*., 2007; Fernandez-Egea *et al*., 2009), although not consistently (Sengupta *et al*., 2008). Differences in subjects and methods could result in different conclusions. So, it needs further studies uniformed in subjects and methods to find out whether different influences on metabolism exist in these medications. In addition, some improvements are expected to demonstrate in future studies including randomized controlled trials, larger sample size and so on.

In summary, some conclusions were obtained from this study. (i) Paliperidone ER, aripiprazole and ziprasidone are effective to treat first-episode schizophrenia, and the ranking of efficacy from high to low is paliperidone ER > aripiprazole > ziprasidone. (ii) Paliperidone ER can impair lipid metabolism potentially and has no influence on glucose metabolism. (iii) Aripiprazole can damage glucose metabolism and has little influence on lipid metabolism. (iv) Ziprasidone is considered an atypical antipsychotic with no harm to glucose and lipid metabolism.

CONFLICTS OF INTEREST

The authors have declared no conflict of interest.

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